RECENT TRENDS IN TECHNOLOGY

Artificial Intelligence

"Artificial Intelligence is the science of making machines do things that would require intelligence if done by humans." **John McCarthy(Father of AI)**

Artificial intelligence (AI) is the capability of computational systems to perform tasks typically associated with human intelligence, such as learning, reasoning, problem-solving, perception, and decision-making.

It is the field of research in computer science that involves making machines do the tasks that are traditionally considered to be possible only for Humans. It deals with making machines to perceive their environment and use learning and intelligence to take actions that maximize their chances of achieving defined goals.

Scope of AI

- ★ Robotics
- ★ Search
- ★ Planning
- ★ Machine learning
- ★ Image processing
- ★ Expert systems
- ★ Natural Language Processing

Asimov's Laws of Robotics

Laws of Robotics are a fictional, foundational code established in 1942 designed by science fiction Author Issac Asimov for his science fiction series "I, Robot" to ensure safety in human-robot interactions.

These laws are:

1. A robot may not injure a human being or, through inaction, allow a human being to come to harm.
2. A robot must obey the orders given it by human beings except where such orders would conflict with the First Law.
3. A robot must protect its own existence as long as such protection does not conflict with the First or Second Law.

The precedence of these laws follow a top-down precedence guided by the meta law which ensures a lower to higher precedence.

APPLICATION Areas OF AI

Artificial intelligence (AI) applications are software programs that use AI techniques to perform specific tasks. These tasks can range from simple, repetitive tasks to complex, cognitive tasks that require human-like intelligence.

AI applications are becoming increasingly common in a wide variety of industries, including healthcare, finance, retail, and manufacturing. As AI technology continues to develop, we can expect to see even more innovative and groundbreaking AI applications in the future.

AI in business intelligence

AI is playing an increasingly important role in business intelligence (BI). AI-powered BI tools can help businesses collect, analyze, and visualize data more efficiently and effectively. This can lead to improved decision-making, increased productivity, and reduced costs.

Some of the ways that AI is being used in Business Intelligence include:

- **Data collection:** Collecting data from a variety of sources, including structured data (for example, databases) and unstructured data (for example, text documents, images, and videos)
- **Data analysis:** To analyze data and identify patterns, trends, and relationships
- **Data visualization:** AI can help create visualizations that make it easier to understand data
- **Decision-making:** Insights and recommendations generated by AI models can help drive data-driven decision-making for businesses

AI in healthcare

AI is also playing an increasingly important role in healthcare. AI-powered tools can help doctors diagnose diseases, develop new treatments, and provide personalized care to patients. For example:

- **Disease diagnosis:** AI can be used to analyze patient data and identify patterns that may indicate a disease. This can help doctors diagnose diseases earlier and more accurately.
- **Treatment development:** By analyzing large datasets of patient data, AI can identify new patterns and relationships that can be used to develop new drugs and therapies.
- **Personalized care:** By analyzing a patient's data, AI can help doctors develop treatment plans that are tailored to the patient's specific needs.

AI in education

AI could be used in education to personalize learning, improve student engagement, and automate administrative tasks for schools and other organizations.

- **Personalized learning:** AI can be used to create personalized learning experiences for students. By tracking each student's progress, AI can identify areas where the student needs additional support and provide targeted instruction.

- **Improved student engagement:** AI can be used to improve student engagement by providing interactive and engaging learning experiences. For example, AI-powered applications can provide students with real-time feedback and support.
- **Automated administrative tasks:** Administrative tasks, such as grading papers and scheduling classes, can be assisted by AI models, which will help free up teachers' time to focus on teaching.

AI in finance

AI can help financial services institutions in five general areas: personalize services and products, create opportunities, manage risk and fraud, enable transparency and compliance, and automate operations and reduce costs. For example:

- **Risk and fraud detection:** Detect suspicious, potential money laundering activity faster and more precisely with AI
- **Personalized recommendations:** Deliver highly personalized recommendations for financial products and services, such as investment advice or banking offers, based on customer journeys, peer interactions, risk preferences, and financial goals
- **Document processing:** Extract structured and unstructured data from documents and analyze, search, and store this data for document-extensive processes, such as loan servicing, and investment opportunity discovery

AI in Entertainment

AI has transformed the entertainment landscape.

Personalized Recommendations

Streaming services use AI to ensure users find content they love without endless scrolling.

- Content Discovery: Platforms like Netflix and Spotify analyze viewing history, skip rates, and even the time of day to suggest movies or songs.

- Dynamic Artwork: AI can change the thumbnail of a movie to match a user's specific interests (e.g., showing a romantic scene to a fan of dramas).

Content Creation and Production

AI acts as a co-creator, helping artists push the boundaries of traditional media.

- Scriptwriting and Storyboarding: AI tools can analyze successful plot structures to help writers brainstorm dialogue or generate visual storyboards before filming begins.

- Music Composition: Algorithms can generate original background scores for video games and films, adapting the "mood" of the music in real-time based on the action on screen.

- Visual Effects (VFX): AI simplifies labor-intensive tasks like de-aging actors, removing unwanted objects from frames, and creating realistic CGI environments.

Gaming and Interactive Media

In the gaming world, AI creates living, breathing environments that react to player choices.

- Smarter NPCs: Non-playable characters (NPCs) use AI to exhibit more human-like behavior, making interactions feel more natural and unpredictable.
- Procedural Generation: AI can automatically generate vast, unique game worlds (like in No Man's Sky), ensuring that no two players have the exact same experience.

Post-Production and Accessibility

AI makes the final stages of production faster and more inclusive.

- Automated Editing: AI can scan hours of raw footage to identify key moments, helping editors quickly create trailers or highlight reels.
- Localization: AI-powered dubbing and subtitling allow content to be translated into dozens of languages almost instantly, making global releases much easier.

AI in Manufacturing

Some ways that AI may be used in manufacturing include:

- Improved efficiency: Automating tasks, such as assembly and inspection
- Increased productivity: Optimizing production processes
- Improved quality: AI can be used to detect defects and improve quality control

Additional AI applications

In addition to the applications listed above, AI is also being used in a variety of other industries, including:

- **Retail:** AI is being used to personalize the shopping experience, recommend products, and manage inventory
- **Transportation:** AI is being used to develop self-driving cars and improve traffic management.
- **Energy:** AI is being used to improve energy efficiency and predict energy demand.
- **Government:** AI is being used to improve public safety, detect crime, and provide citizen services.

Opportunities Associated with AI

- Complex Problem Solving and Higher Computational Ability
- Democratization of Knowledge
- Transformation of Human labor and Increased productivity
- Use of Pattern recognition for disaster management, climate change mitigation, etc.
- Creation of New jobs and new markets
- Excels at being human’s copilot i.e. assistant to humans

Limitations of AI

- Produce Hallucinations and Confident Errors
- Lack of transparency due to private control and profit motive
- Lack of Intelligence (emotional and general)
- Lack of True Common sense
- High Cost of Resources
- Data dependence

Threats Associated with AI

- Privacy Threats
- Deepfakes and Misinformation
- Algorithmic Bias towards the status quo
- Replacement of Old Jobs
- Ease of Mass surveillance
- Increases Cybersecurity threats

Turing Test in AI

Alan Turing in his 1950 paper "Computing Machinery and Intelligence," proposed a test called ***Imitation Game*** and it is the most famous benchmark for determining if a machine can "think."

The test involves three participants in a controlled environment to ensure the judge cannot see or hear the others:

1. **The Interrogator (Judge):** A human who asks questions via a text-based interface.
2. **The Human Respondent:** A person who answers honestly to prove they are human.
3. **The AI Respondent:** A machine programmed to mimic human conversation and "trick" the judge.

Natural Language Processing

Natural Language Processing (NLP) is a branch of artificial intelligence that enables computers to understand, interpret, generate, and process human languages (text and speech).

It uses linguistics with machine learning and deep learning models to perform tasks like machine translation, sentiment analysis, chatbots, and voice assistants. It enables machines to read, decipher, and make sense of human languages in a way that is valuable.

Approaches for Natural Language Processing

Symbolic Approach [Hard Coded Logic]

The Symbolic Approach was the foundational era of NLP, relying on human experts to manually code thousands of linguistic rules. In this paradigm, a computer processes language by following strict ***"if-then" logic***—for example, a rule might state that if a sentence starts with "Who," it is likely a question.

This approach is highly transparent and precise for structured tasks, it is incredibly rigid. It fails to handle the messy reality of human speech, such as slang, typos, or sarcasm, because it cannot "understand" anything that hasn't been explicitly defined in its pre-programmed dictionary.

Statistical Approach

Statistical approach in NLP automatically extracts, classifies and labels elements of text and voice data and then assigns a statistical likelihood to each possible meaning of those elements. This relies on machine learning, enabling a sophisticated breakdown of linguistics such as part-of-speech tagging.

Statistical NLP introduced the essential technique of mapping language elements, such as words and grammatical rules to a vector representation so that language can be modeled by using mathematical (statistical) methods, including regression or Markov models.

Neural Networks

Neural networks are computational models inspired by the human brain, using interconnected layers of artificial neurons to process data, learn complex patterns, and make predictions. They consist of input, hidden, and output layers that transform data, often powering AI applications like computer vision and natural language processing.

Robotics

Robotics is the interdisciplinary study and practice of the design, construction, operation, and use of robots. Robotics usually combines four aspects of design work: *a power source (e.g. a battery), mechanical construction, a control system (electrical circuits), and software (run by remote control or artificial intelligence)*.

Design of a Robot

Robotics usually combines four aspects of design work to create a robot:

- **Power source:** Potential energy sources include wired electricity, a battery, and/or petrol.
- **Mechanical construction:** A physical form or combination of forms is designed to functionally achieve tasks within a given range of environments. This can include locomotive elements such as wheels and caterpillar tracks, as well as hydraulic limbs and manipulators (e.g. hands).
- **Control system:** Electrical circuits (utilizing components such as diodes and transistors) are used to run software, govern motor movement and read sensors.
- **Software:** A program is how a robot decides when or how to do something. Robotic programs can be run by remote control, artificial intelligence (AI), or a hybrid of the two. AI programming is an important part of robotic navigation.

Sensors and Actuators

Sensors and actuators are critical components in automation, robotics, and IoT, forming the input-output loop between systems and their environment. Sensors detect physical, chemical, or biological properties (temperature, pressure, light) and convert them into electrical signals.

Actuators receive these signals and convert energy (electrical, hydraulic, pneumatic) into physical movement or action.

Sensor

A sensor is a device used for the conversion of physical events or characteristics into electrical signals. This is a hardware device that takes the input from the environment and gives it to the system by converting it. For example, a thermometer takes the temperature as a physical characteristic and then converts it into electrical signals for the system.

Types of Sensors

- ➔ Temperature Sensors: Take temperatures.
- ➔ Light Sensors: Light intensity sensors: It has the function of detecting the intensity of the light.
- ➔ Pressure Sensors: To use it to measure pressure in gases or liquids.
- ➔ Motion Sensors: Recognize motion in an established region.

Advantages of Sensors

- ➔ Offer timely and accurate information as this is a critical requirement by the high release frequency.
- ➔ Support automation and management of systems.
- ➔ Improve safety by maintaining checks on important parameters.

Disadvantages of Sensors

- Sometimes it can be costly, particularly the high precision sensors.
- It can sometimes need some adjustments and can also probably require maintenance in the long run.
- Interference from the environment is easy in this process.

Actuator

An actuator is a device that converts the electrical signals into physical events or characteristics. It takes the input from the system and gives output to the environment. For example, motors and heaters are some of the commonly used actuators.

Types of Actuators

- ➔ Linear Actuators: Utilize a linear motion to convert energy, Kinetic/pendulum.
- ➔ Rotary Actuators: This will affect the creation of rotational motion.
- ➔ Hydraulic Actuators: How does fluid power give motion.
- ➔ Pneumatic Actuators: Function with use of compressed air.

Advantages of Actuators

- ➔ Assist in providing a fine level of control of mechanical installations.
- ➔ They should enable automation and therefore minimize the need for intervention of human participants.

- ➔ Available in a range of variations and suitability in multiple operations ranging from everyday uses to industrial use.

Disadvantages of Actuators

- ➔ May consume much power in its operation particularly when used in places that involve much power such as in large industries.
- ➔ May be large and costly to both install and maintain.
- ➔ As a disadvantage there is a circumstance that, with time the component is liable to mechanical wear and tear.

Applications of Robotics

Robotics has moved far beyond simple "arm-on-a-track" automation. In 2026, the field is dominated by Physical AI, where robots use advanced sensors and real-time reasoning to interact with unpredictable environments.

Here are the primary application areas of modern robotics:

1. Manufacturing & Logistics (Industry 4.0)

This remains the largest sector, but the focus has shifted from "caged" robots to collaborative systems.

- Cobots (Collaborative Robots): Unlike traditional robots, cobots work alongside humans without safety fences. They are used for delicate assembly, machine tending, and packaging.
- AMRs (Autonomous Mobile Robots): These robots navigate warehouses independently using LiDAR and 3D vision, moving pallets and picking orders without needing floor tracks or magnets.
- "Dark Factories" (Lights-out Manufacturing): Some facilities now run fully autonomous "night shifts," where robots handle the entire production line with zero human intervention until the morning crew arrives.

2. Healthcare & Life Sciences

Robotics is increasing precision and extending the reach of medical professionals.

- **Surgical Assistance:** Systems like the *da Vinci* (and its 2026 competitors) allow for micrometer-precision in neurosurgery and laparoscopy, reducing recovery times by up to **20%**.
- **Exoskeletons:** Wearable robotic suits are used in rehabilitation to help paralyzed patients walk or to assist elderly individuals with mobility.
- **Disinfection & Logistics:** Autonomous robots roam hospital halls to disinfect rooms using UV light or to deliver medication and linens to nursing stations.

3. Agriculture (AgTech)

Faced with labor shortages, the farming industry has turned to "Smart Agriculture."

- **Autonomous Tractors:** These use GPS and AI to plant seeds, till soil, and apply fertilizer with zero human drivers.
- **Precision Harvesting:** Soft-robotics grippers allow robots to pick fragile fruits (like strawberries or tomatoes) without bruising them—a task that was historically impossible for machines.
- **Drones & Monitoring:** Swarms of drones monitor crop health via thermal imaging, identifying pest outbreaks or irrigation leaks before they spread.

4. Emerging Humanoid Applications

2026 has seen the first real-world deployments of general-purpose humanoid robots in 'human spaces.'

- **Labor Gap Filling:** Humanoids are being trialed in automotive plants (like BMW and Hyundai) for "brownfield" tasks—jobs in existing factories designed for human proportions, such as moving bins or simple part fitting.
- **The "Gig Economy" for Robots:** In some regions, humanoids are now rented out for short-term events, acting as greeters, performers, or interactive guides at trade shows and corporate gatherings.

5. Hazardous Environments & Exploration

Robots go where humans cannot survive.

- **Space Exploration:** Robotic "avatars" and rovers are currently being used to build the groundwork for lunar bases and to collect samples on Mars.
- **Disaster Response:** Quadruped robots (often called "robotic dogs") are deployed into collapsed buildings or chemical leak sites to map the area and search for survivors using thermal sensors.

Cloud Computing

Cloud computing is a way of accessing compute and storage systems without actually owning and doing active management of the resources. In today's world computer and storage demands are very dynamic hence purchasing, maintaining and upgrading systems could be a huge investment of time and money.

Companies like AWS (Amazon Web Services), Microsoft Azure, Google Cloud Platform (GCP) provide compute and storage servers on demand and charge for what you use. These cloud services can be used to host static websites, e-commerce stores, company's internal data, etc. It has proven to be extremely useful for startups where compute resources can vary largely over time.

Cloud Computing Service Models

SaaS (Software as a Service)

Software as a Service (SaaS) is a cloud-based model delivering software applications over the internet via subscriptions, eliminating the need for local installation or maintenance. Providers manage infrastructure, security, and updates, making it highly accessible and scalable. Common examples include Gmail, Salesforce, Dropbox, and Netflix.

PaaS (Platform as a Service)

Platform as a Service (PaaS) is a cloud computing model providing a complete, managed environment—including infrastructure, operating systems, databases, and development tools—for developers to build, test, and deploy applications without managing underlying servers. It accelerates development cycles and reduces costs by handling infrastructure maintenance, allowing focus on code.

IaaS (Infrastructure as a Service)

Infrastructure as a Service (IaaS) is a cloud computing model providing virtualized, on-demand computing resources—servers, storage, and networking—over the internet on a pay-as-you-go basis. It allows businesses to avoid purchasing, managing, and maintaining physical, on-premises data center hardware. Users manage operating systems and applications, while providers manage the underlying infrastructure.

Cloud Computing Deployment Models

Public Cloud

- ➔ Service Provider makes resources such as Compute, storage, and applications available to the general public over the internet.
- ➔ Any user can log in and use these services.
- ➔ You pay for the number of resources you use.
- ➔ Users have lesser control over their data.

Private Cloud

- ➔ The vendor offers hosted services to fewer users with firewall security.
- ➔ Private cloud minimizes security issues.
- ➔ It provides greater control over the data.
- ➔ Typically used by organizations with a focus on data security.

Hybrid Cloud

- ➔ Hybrid cloud computing as the name implies uses a combination of private and public cloud services.

- ➔ Certain services are hosted with private cloud while others with the public cloud.
- ➔ With hybrid cloud service, enterprises can keep crucial data into private space and other data into public space thus leveraging the best of both worlds.

Advantages of Cloud Computing

- ➔ **Cost Efficiency:** It eliminates high upfront capital expenses (CapEx) by shifting to a "pay-as-you-go" operational model (OpEx).
- ➔ **Infinite Scalability:** You can instantly increase or decrease computing power and storage to match fluctuating business demands, which gives theoretically infinite scalability.
- ➔ **Global Accessibility:** Employees can securely access data and applications from any device, anywhere in the world with an internet connection.
- ➔ **Enhanced Collaboration:** Shared cloud environments allow multiple users to edit documents and sync data in real-time, regardless of their location.
- ➔ **Automatic Software Updates:** Cloud providers handle all system and security maintenance, ensuring you always use the latest, most secure versions.
- ➔ **Disaster Recovery:** Geographically distributed data centers provide robust backup and recovery solutions that protect against local hardware failures or natural disasters.
- ➔ **Speed to Market:** New IT resources can be provisioned in minutes rather than weeks, allowing businesses to launch products faster.
- ➔ **Advanced Security:** Top-tier providers invest billions in specialized security measures like AI-driven threat detection and high-level encryption that most small firms cannot afford.
- ➔ **Environmental Sustainability:** By sharing centralized resources, cloud computing significantly reduces energy waste and lowers the overall carbon footprint compared to on-premise servers.
- ➔ **Data Loss Prevention:** Centralized storage prevents critical information from being lost if a physical laptop or local hard drive is stolen or damaged.

Disadvantages of Cloud Computing

- ➔ **Loss of Control** – When using cloud services, you hand over control of your data and applications to the provider, limiting your ability to manage systems directly.
- ➔ **Service Disruptions** – Cyber-attacks, power failures, or internet outages at the provider's end can cause unexpected downtime and interrupt business operations.
- ➔ **Provider Shutdown Risk** – If a cloud provider goes out of business or changes its services, your operations may be severely affected until a new provider is found.

- **Security Threats** – Storing sensitive data on the cloud increases the risk of hacking, data breaches, and cyber threats beyond your direct control.
- **Vendor Lock-in** – Migrating from one cloud provider to another can be complex, costly, and technically challenging due to differences in platforms and configurations.
- **Wrong Provider Selection** – Choosing an unsuitable cloud provider may result in poor service quality, missing features, and operational inefficiencies.
- **Inadequate Customer Support** – Some cloud providers offer limited technical support, making it difficult for non-technical users to resolve issues quickly.
- **Limited Internet Dependency** – Cloud computing heavily depends on stable internet connectivity, and slow or unstable networks can reduce performance.
- **Hidden Costs** – Additional charges for storage, bandwidth, upgrades, or support services can increase overall expenses unexpectedly.
- **Compliance and Legal Issues** – Storing data in different geographical locations may create legal, regulatory, or compliance challenges for businesses.

Big Data

Big data refers to massive, complex, and rapidly growing data sets (structured, unstructured, and semi-structured) that exceed the capacity of traditional data processing systems. It is defined by high volume, velocity, variety, veracity, and value (5 Vs), often stored in clouds or data lakes, used for predictive analytics, machine learning, and AI.

The need of Big Data

Big data is preferred over traditional databases when handling massive, high-velocity, and unstructured data (petabytes+), which conventional relational systems cannot efficiently process. It offers superior scalability, advanced analytics, and the ability to analyze varied data types like video, social media, and logs for real-time insights.

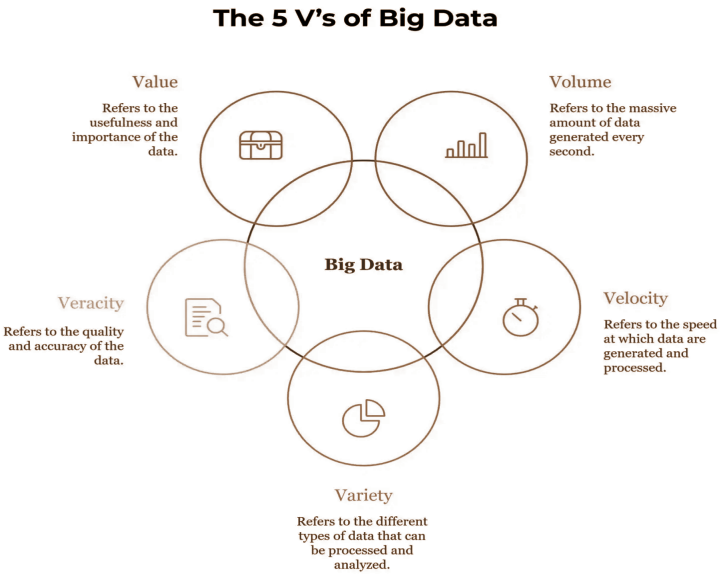
Big Users: Not that long ago, 1,000 daily users of an application was a lot and 10,000 was an extreme case. Today, with the growth in global Internet use, the increased number of hour's users spend online, and the growing popularity of smartphones and tablets, it's not uncommon for apps to have millions of users a day.

Big Data: Data is becoming easier to capture and access through third parties such as Facebook, D&B, and others. Personal user information, geo location data, social graphs, user-generated content, machine logging data, and sensor-generated data are just a few examples of the ever-expanding array of data being captured.

Cloud Computing: Today, most new applications (both consumer and business) use a three-tier Internet architecture, run in a public or private cloud, and support large numbers of users.

5Vs of Bigdata

Volume



"Volume" refers to the high number of data points in a big data set. Streaming services like Netflix or YouTube are great examples of this. These platforms cater to millions of users who stream videos, which generates an enormous amount of data. Netflix not only has to store this colossal volume of streaming data but also user preferences, search histories, and interactions.

The volume of data generated helps Netflix use sophisticated algorithms to recommend shows and movies, leading to a more personalized user experience. While the large volume of data generated leads to more targeted consumer recommendations, analyzing and managing this information requires advanced storage and processing capabilities.

Veracity

The term "Veracity" refers to the trustworthiness and quality of the data. With such a high volume of data generated daily, it remains a challenge to ensure the data you work with is unbiased and correctly represents what it's supposed to. In this case, verifying and validating your data at each step of the collection and analysis process is imperative.

Depending on the nature of the data, missing values, noise, model approximation, ambiguity, and bias can influence the veracity of the data. The acceptable veracity of the data will depend on what type of data you have and what you're trying to achieve. For example, when it comes to medical data, the veracity of the data acceptable for research purposes is much different than the veracity for clinician decisions.

Velocity

"Velocity" in big data refers to the speed of data generation and how fast professionals can collect and process it. This varies depending on the data source. For example, millions of posts are sent out daily on social media sites like X, while wearables like Apple Watches collect health metrics continuously.

However, velocity is not just about the rapid arrival rate. In many fields, professionals make snap decisions as the data arrives. For example, financial institutions trading in the stock market use high-velocity data to make split-second decisions that could involve millions of dollars.

Variety

"Variety" refers to the wide range of data types and sources in a big data set, including structured and unstructured data.

Structured data includes more defined data types, such as databases of names and numbers. Unstructured data, on the other hand, includes data types like text, audio, images, and social media posts. Semi-structured data is a mix of the two. For example, in health care, patient data may include structured records such as age, diagnosis, and treatment history, and unstructured data such as medical notes, health images, and even genetic information.

When dealing with variety in big data, you must aggregate and analyze your data to preserve its meaning while also driving more impactful insights. This requires you to employ complex processing techniques and advanced analytics. While single data points may have biases, a benefit of the variety in big data is that you get multiple reference points on the topic of interest to build a better picture.

Value

The "Value" of big data comes from the insights and patterns you can find. Because big data incorporates data from diverse sources and formats, you can build insights into metrics of interest, such as customer behavior, how the market moves, business performance, and more. For instance, while structured data can reveal numerical trends and patterns, unstructured text data from social media posts or customer reviews can uncover sentiments and opinions that drive human behavior.

Who has Big Data?

Big Data is primarily held and utilized by major technology, finance, and logistics organizations to drive decision-making and innovation. Top entities include tech giants like Google, Amazon, Microsoft, IBM, Oracle, and Meta. Other major holders include data centers like China Mobile and China Telecom, as well as companies like eBay, Netflix, and Marriott.

These organizations use big data technologies like Apache Hadoop, Spark, and Kafka to analyze vast datasets for personalized marketing, dynamic pricing, and operational efficiency.

Data Brokers and Bigdata

Data brokers are companies that aggregate, analyze, and sell massive, "big data" profiles on billions of individuals, sourced from social media, public records, and online activity. They create detailed consumer categories—like "Affluent Baby Boomers"—used for targeted marketing, risk assessment, and surveillance capitalism, often without direct consumer knowledge or consent.

What Data Brokers Do with Big Data?

- **Data Aggregation:** They collect vast, diverse datasets (volume, velocity, variety) including demographics, financial data, health information, and location tracking.
- **Analytics and Profiling:** Brokers use machine learning and proprietary algorithms to analyze behavioral patterns and build detailed, predictive profiles.
- **Monetization:** These profiles are sold to marketers, employers, and financial institutions to predict behavior, assess risk, and target advertising.

Advantages of Big Data

- ➔ **Improved Decision-Making:** Organizations can make faster, more informed, and strategic decisions by analyzing large, diverse data sets.
- ➔ **Enhanced Customer Experience:** Companies can deeply understand customer behavior, needs, and preferences, allowing for personalized marketing and improved satisfaction.
- ➔ **Operational Efficiency and Cost Reduction:** By optimizing production, analyzing feedback, and anticipating equipment downtime with predictive analytics, businesses can reduce maintenance costs and streamline operations.
- ➔ **Fraud Detection and Risk Management:** Real-time analysis of transaction data enables financial institutions and other businesses to detect anomalies, prevent fraud, and mitigate risks.
- ➔ **Competitive Advantage and Innovation:** Companies can identify market trends early, develop new products faster, and gain a competitive edge.
- ➔ **Improved Data Storage and Access:** Modern big data storage provides faster, more reliable access, improved backup solutions, and lower overall costs.

Threats Associated with Big Data

Beyond the technical and financial hurdles, Big Data presents significant societal risks related to control and influence.

- ➔ **Mass Surveillance:** Governments and corporations can monitor movements, communications, and behaviors in real-time, leading to a "panopticon" effect where privacy is virtually non-existent.
- ➔ **Behavioral Manipulation:** Detailed consumer profiles allow platforms to use "nudges" and psychological triggers to influence everything from purchasing habits to lifestyle choices.
- ➔ **Political Micro-targeting:** Data is used to create "echo chambers," delivering highly personalized (and sometimes inflammatory) political messages that can polarize societies and influence elections.
- ➔ **Predictive Policing:** Using historical data to predict crime can result in the over-policing of specific neighborhoods, reinforcing systemic prejudices rather than preventing actual crime.
- ➔ **Social Engineering:** Aggregated data makes it easier for bad actors to craft highly convincing phishing attacks or "deepfakes" designed to deceive specific individuals.
- ➔ **Erosion of Autonomy:** As algorithms begin to recommend what we read, who we date, and where we work, human agency and the ability to make spontaneous, uninfluenced choices are diminished.

- ➔ **Data Persistence:** Information on the internet is nearly impossible to delete; a mistake or a piece of data from a person's youth can haunt their professional and personal life forever.
- ➔ **Exploitation of Vulnerabilities:** Companies can identify and target individuals during moments of weakness (e.g., targeting gambling ads to those showing signs of addiction or financial stress).
- ➔ **Information Asymmetry:** A massive power gap is created between the "data giants" who own the information and the individuals who generate it, leaving the public with little leverage.

Structured, Semi Structured and Unstructured Data

Structured data

It is the most traditional and highly organized form of information. It resides in fixed fields within a record or file, typically following a rigid schema (a predefined blueprint). This data is most often stored in relational databases (SQL) and is arranged in rows and columns, making it extremely easy for computer programs to search, sort, and query.

Because it is so predictable, think of dates, zip codes, or names, it requires very little processing before it can be used for business intelligence. However, its main disadvantage is its lack of flexibility; if the requirements of the data change, the entire database structure often needs to be redesigned.

Semi-Structured Data

Semi-structured data occupies the middle ground between organized tables and raw files. While it does not reside in a rigid relational database, it contains internal tags, markers, or metadata that separate data elements and establish a hierarchy. This allows the data to be "self-describing."

For example, an email is semi-structured: while the body of the message is free-form text, the metadata (sender, recipient, and time) provides enough structure to be indexed. Common formats include JSON and XML.

The primary advantage of semi-structured data is its flexibility; it can accommodate changes in data types more easily than structured data, which is essential for modern web applications and APIs.

Unstructured Data

unstructured data is defined by its lack of a predefined data model or organizational framework, making it incompatible with traditional relational databases. This data type includes a wide range of formats, including high-resolution video, raw sensor logs, satellite imagery, and long-form textual documents. Because it exists in a raw state without structural markers, it is inherently difficult to search or index using conventional algorithms.

Organizations must therefore leverage advanced Natural Language Processing (NLP) and Machine Learning to identify meaningful patterns or extract actionable intelligence. Despite its complexity, unstructured data is considered the most valuable source for uncovering nuanced consumer insights and deep operational trends.

NO-SQL(NOT ONLY SQL)

NoSQL stands for "Not Only SQL". It refers to a category of database management systems designed to handle the massive volume, social variety, and high velocity of Big Data. Unlike traditional relational databases that require rigid tables and predefined schemas, NoSQL databases utilize flexible data models such as document, graph, key-value, and wide-column stores. This flexibility allows developers to store unstructured and semi-structured data—like social media posts or sensor logs—without first establishing a fixed format.

One of the primary advantages of NoSQL is its ability to scale horizontally. Instead of upgrading a single expensive server, organizations can distribute data across thousands of commodity servers, a process known as sharding. While they may sacrifice some of the strict transactional consistency found in SQL, NoSQL systems like MongoDB, Cassandra, and Amazon DynamoDB prioritize high availability and performance. This makes them the foundational technology for real-time web applications and large-scale data processing.

Big Data in the age of AI

Big Data and Artificial intelligence (AI) are very closely interrelated. Big Data provides a huge amount of information, and AI uses that information to learn and make smart decisions. Without data, AI cannot improve or work properly. At the same time, if we only collect data but do not analyze it using AI, the data will not be very useful.

Both Big Data and AI dependent on each other. Big Data gives the input, and AI gives the output in the form of predictions, recommendations, or solutions. For example, when we use social media or online shopping apps, our data is collected, and AI studies it to suggest content or products that match our interests.

How AI uses Big Data?

AI systems analyze Big Data to:

- ➔ Detect hidden patterns
- ➔ Make predictions
- ➔ Automate decisions
- ➔ Personalize user experiences
- ➔ Detect fraud and anomalies

For example:

- Netflix uses AI to recommend movies based on viewing data.
- Amazon analyzes purchase data for product recommendations.
- Google uses AI to improve search results and ads targeting.

Ecommerce

E-commerce (electronic commerce) is the buying and selling of goods and services, or the transmitting of funds or data, over an electronic network, primarily the internet. Ecommerce works by connecting sellers with customers and allowing exchanges to take place online. It can work in many different ways and take many forms.

E-Commerce Process Overview

- ➔ The seller chooses online selling channels, like a website or social media, and promotes products or services for sale.
- ➔ Customers find the products or services and place orders.
- ➔ A payment processor enables the exchange of the goods or services electronically via payment options like credit cards or digital currencies.
- ➔ The customer receives a confirmation email or SMS along with a printable receipt.
- ➔ If the transaction is for goods, the seller ships the products and sends the customer a tracking number via email or SMS. If the transaction is for a service, the service provider can reach out to schedule and complete the service.

Ecommerce Business Model (Types)

Business-to-Consumer (B2C)

B2C is the most common type of eCommerce model, where businesses sell directly to consumers. In this model, you are targeting individuals who want to buy products for personal use. B2C typically involves smaller orders and shorter sales cycles compared to B2B (Business-to-Business).

Example: If you run an online store selling clothes or gadgets, you are operating under the B2C model. Customers browse your products, add them to their cart, and purchase them directly through your website.

Business-to-Business (B2B)

B2B eCommerce focuses on transactions between businesses. Businesses sell products or services to other companies rather than individuals. B2B typically involves larger order quantities, long-term contracts, and a more complex sales process.

Example: A company that sells office supplies in bulk to other businesses operates under the B2B model. The focus here is on delivering products that help businesses run smoothly, and transactions are often larger than in B2C.

Consumer-to-Consumer (C2C)

C2C eCommerce involves consumers selling directly to other consumers, typically through online marketplaces like eBay, Craigslist, or Etsy. With this model, individuals list their own products for sale, and buyers purchase from them without a business acting as the middleman.

Example: If you sell handmade jewelry on Etsy or list a used phone on eBay, you're participating in the C2C model. Buyers browse listings, and you, as the seller, fulfill orders directly.

Consumer-to-Business (C2B)

In the C2B model, individuals sell products or services to businesses. This is less common than B2C or B2B but is growing thanks to the rise of freelance services and platforms like Upwork or Fiverr, where businesses hire individual professionals for specific tasks.

Example: If you're a freelance graphic designer offering services to businesses or a blogger writing sponsored content for companies, you're working in the C2B model.

Advantages of E-commerce

24/7 Availability: Unlike a physical storefront, an online shop never sleeps—customers can buy whenever they want to.

Global Reach: You aren't limited by geography; you can sell to someone across the street or across an ocean.

Lower Overhead: No physical rent, utility bills for a showroom, or large in-person staff, which significantly cuts operational costs.

Personalization: Using data, you can show customers products tailored specifically to their tastes, increasing the chance of a sale.

Scalability: It's much easier to add new products or handle a surge in traffic online than it is to expand a physical building.

Niche Marketing: You can find and target "your people" (no matter how specific their interests) much more effectively through search engines and social media.

Direct-to-Consumer (DTC) Sales: Brands can bypass wholesalers and retailers, keeping more of the profit and controlling the brand experience.

Faster Buying Process: Customers can find what they want, click, and pay in seconds without waiting in a checkout line.

Detailed Analytics: You get instant data on what's selling, who's looking, and where people are dropping off, allowing for quick pivots.

Inventory Management: Automated systems can track stock levels in real-time across multiple warehouses, reducing the risk of overselling.

Disadvantages of E-Commerce

Security Risks: Sites are constant targets for hackers, phishing, and credit card fraud, requiring expensive, ongoing protection.

Lack of Physical Touch: Customers can't feel, smell, or try on products, which can lead to hesitation or disappointment upon delivery.

Shipping & Logistics: Managing carriers, dealing with lost packages, and meeting "Prime-speed" expectations is a massive operational burden.

High Return Rates: Online shoppers return items much more frequently than in-store shoppers, eating into profits and doubling logistics work.

Fierce Competition: You aren't just competing with the shop next door; you're competing with global giants and millions of other startups.

Technical Downtime: If your site crashes or the server goes down, your business effectively ceases to exist until it's fixed.

Low Barriers to Entry: Because it's "easy" to start, many markets are oversaturated, making it harder (and more expensive) to stand out.

Customer Trust Issues: New or small brands often struggle to convince skeptical users to hand over their payment information.

Ad Cost Inflation: As more businesses move online, the cost of "buying" traffic through Google or Meta ads continues to climb.

Isolation: You lose the personal touch and immediate relationship-building that happens during a face-to-face transaction.

E-Commerce vs E Business

E-Business involves executing all types of business transactions and services via the web. This includes procuring raw materials, online education, commercial transactions, monetary transactions on the internet and more. This indicates an online presence of all types of businesses and services.

On the other hand E-Commerce is about conducting commercial transactions via the web or getting involved in selling and buying items over the internet. It also covers monetary transactions.

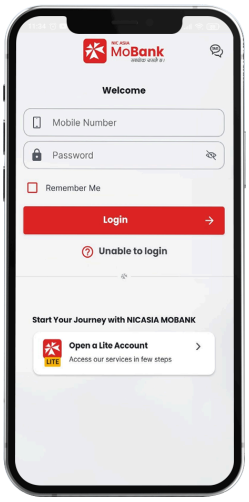
Basically, Ecommerce acts as a subset of E-business. To have an e-business all of the activities relating to a business should have online presence. However, to have e-commerce, the only requirement is the ability to conduct financial transactions online. Some businesses only offer ecommerce while the business being completely offline for the other part. While there are certain businesses that operate completely online. A business must offer e-commerce to be an e-business.

E-Commerce	E-Business
Carrying out commercial transactions online	Conducts all kinds of business activities and services over the internet
Buying/ selling, monetary transactions online	Online presence of the business
A sub-set of E-Business, it is also a narrow concept	A super-set of E-Commerce. Business transactions are supported in E-Business
Limited transactions	Transactions are not limited
Involves use of only one website	Multiple websites and CRMs, ERPs that connect different business processes are used
Mandatory use of internet	Internet, Intranet or Extranet are used
It is more relevant in B2C, Business to customer context	This is more appropriate to B2B or Business to Business context
Also cover external or outward business processes	Covers internal as well as external business activities or processes

M-Commerce

M-commerce (mobile commerce) means buying and selling goods or services through mobile devices like smartphones and tablets. It's basically e-commerce, but done on a phone instead of a desktop computer.

It is a subset of e-commerce that enables on-the-go, instant transactions, including mobile banking, in-app purchases, and digital wallets. It offers convenience through features like mobile apps, mobile payments (e.g., Apple Pay), and personalized shopping experiences.



Mobile Banking

Mobile banking is a service that lets you manage your bank account using a smartphone or tablet. Instead of visiting a branch or using a computer, you use your bank's mobile app.

connect IPS
pay direct from bank

Global Smart Plus
Global IME Bank Limited

Digi Prabhu
F1soft

Mobile Wallets

A mobile wallet is a digital app that lets you store money electronically and make payments using your smartphone. Instead of carrying cash, you pay through your phone.

Mobile Shopping

Mobile shopping means buying products or services using a smartphone or tablet. Instead of visiting a physical store or using a desktop computer, customers shop through mobile apps or mobile websites.

E-Governance

E-Governance (Electronic Governance) refers to the use of information and communication technology (ICT) by government institutions to provide services, exchange information, and interact with citizens, businesses, and other arms of government efficiently and transparently.

Objectives of E-Governance

- **Improving Service Delivery** – Faster and easier access to government services for citizens.
- **Transparency** – Reducing corruption by making government processes more open.
- **Efficiency** – Streamlining administrative processes using technology.
- **Citizen Participation** – Engaging citizens in decision-making through digital platforms.
- **Cost Reduction** – Reducing paperwork, travel, and operational costs for government services.

Advantages of E-Governance

- Convenient access to services 24/7.
- Reduced paperwork and bureaucracy.
- Faster decision-making and service delivery.
- Enhanced transparency and accountability.

- ➔ Better resource management and data-driven planning

Challenges to E-Governance

- ➔ Digital divide and lack of ICT literacy among citizens.
- ➔ High initial cost of infrastructure and maintenance.
- ➔ Data security and privacy concerns.
- ➔ Resistance to change within government departments.
- ➔ Dependence on reliable internet connectivity.

Components of E-Governance

Components	Description	Examples
Government to Citizen (G2C)	Digital interaction between government and citizens to provide services and information.	Online tax filing, utility bill payment, e-health services, online application for certificates (birth, death, caste).
Government to Business (G2B)	Services and communication between government and businesses, helping in compliance, licensing, and trade.	E-procurement, online business registration, digital tax payments, import/export licenses.
Government to Government (G2G)	Digital communication and data exchange between government departments to improve coordination and efficiency.	Sharing of land records between departments, inter-department data exchange, joint projects management.
Government to Employees (G2E)	ICT-based services for government employees, including HR management, payroll, training, and internal communication.	Online payroll systems, leave management portals, e-learning platforms, staff notifications.

Telemedicine (E-Medicine)

Telemedicine or E-Medicine is the delivery of healthcare services using electronic communication and information technologies. It allows medical professionals to evaluate, diagnose, and treat patients remotely without requiring physical visits to healthcare facilities.

Objectives of Telemedicine

- ➔ **Access to Healthcare** – Provide medical services to patients in remote or underserved areas.
- ➔ **Time Efficiency** – Reduce the need for travel and waiting times for patients.
- ➔ **Cost Reduction** – Minimize expenses for both healthcare providers and patients.
- ➔ **Specialist Consultation** – Enable access to specialists who may not be locally available.
- ➔ **Health Monitoring** – Continuous monitoring of patients using wearable devices and mobile health applications.

E-Learning and Distant Learning

E-Learning : Learning conducted through electronic media, typically via the internet, computers, or mobile devices. It allows students to access educational content anytime and anywhere.

Distance Learning : A method of education where students and teachers are physically separated. It can use traditional mail, radio, television, or modern digital platforms. E-Learning is a subset of Distance Learning when conducted online.

Objectives

- **Accessibility :** Provide education to learners in remote areas or with mobility constraints.
- **Flexibility :** Allow learners to study at their own pace and schedule.
- **Cost Efficiency** : Reduce travel, infrastructure, and classroom expenses.
- **Resource Availability :** Offer access to a wide range of learning materials and expert instructors.
- **Lifelong Learning :** Support continuous education for professionals and adult learners.

Internet of Things (IOT)

The Internet of Things (IoT) refers to a network of physical devices, vehicles, appliances, and other physical objects that are embedded with sensors, software, and network connectivity, allowing them to collect and share data.

Tasks performed using IOT

IoT enables these smart devices to communicate with each other and with other internet-enabled devices. Like smartphones and gateways, creating a vast network of interconnected devices that can exchange data and perform various tasks autonomously.

- ➔ monitoring environmental conditions in farms.

- managing traffic patterns with smart cars and other smart automotive devices
- controlling machines and processes in factories
- tracking inventory and shipments in warehouses

Advantages of IOT

Predictive Maintenance: Sensors detect equipment issues before they cause costly breakdowns.

Cost Reduction: Automation and optimized resource usage significantly lower operational expenses.

Real-time Tracking: Instant visibility into the location and status of inventory or assets.

Enhanced Productivity: Streamlined workflows allow machines and people to work more efficiently.

Data-Driven Decisions: Continuous data streams provide precise insights for smarter business moves.

Improved Safety: Monitoring hazardous environments remotely protects workers from potential harm.

Resource Optimization: Smart systems minimize waste of electricity, water, and raw materials.

Better Quality of Life: Remote health monitoring and smart home features increase comfort and care.

Disadvantages of IOT

Security Vulnerabilities: Every connected device acts as a potential entry point for hackers and cyberattacks.

Privacy Concerns: Constant data collection leads to personal information being tracked and potentially sold.

High Initial Costs: Implementing hardware, software, and infrastructure requires a massive upfront investment.

System Complexity: Managing thousands of interconnected devices from different brands creates technical chaos.

Internet Dependency: If the network goes down, "smart" devices often lose functionality or become useless.

Job Displacement: Automation of monitoring and manual tasks can lead to the loss of entry-level roles.

Data Overload: The sheer volume of incoming data can overwhelm systems if not properly analyzed.

Electronic Waste: The rapid lifecycle of sensors and gadgets contributes to growing environmental pollution.

APPLICATIONS OF IOT

Industrial & Manufacturing (IIoT)

- **Predictive Maintenance:** Sensors monitor machinery health to schedule repairs before a breakdown occurs.
- **Digital Twins:** Creating virtual replicas of factories to simulate and optimize production workflows safely.
- **Smart Inventory:** RFID and IoT sensors automate stock tracking and reordering to prevent supply chain gaps.

Healthcare (IoMT)

- **Remote Patient Monitoring:** Wearables track vitals like heart rate and glucose, alerting doctors to emergencies instantly.
- **Smart Hospitals:** Tracking the real-time location of medical equipment (ventilators, wheelchairs) and monitoring bed occupancy.
- **Chronic Disease Management:** Connected insulin pumps and cardiac patches provide continuous, automated care.

Smart Cities & Infrastructure

- **Smart Grids:** Automatically balancing energy load and integrating renewable sources like solar and wind efficiently.
- **Intelligent Traffic Management:** Sensors adjust traffic light timing in real-time to reduce congestion and emergency response times.
- **Waste Management:** Smart bins notify collection services only when they are full, optimizing fuel and labor.

Smart Agriculture

- **Precision Farming:** Soil sensors trigger automated irrigation and fertilization only when and where needed.
- **Livestock Tracking:** Wearables monitor animal health and location to detect illness or distress early.
- **Climate Monitoring:** Drones and sensors provide hyper-local weather data to protect high-value crops.

Automotive & Logistics

- **Connected Vehicles (V2X):** Cars communicate with traffic lights and other vehicles to prevent accidents and optimize routes.
- **Cold Chain Monitoring:** Ensuring food and medicine stay at precise temperatures during global transit.

- **Fleet Telematics:** Monitoring driver behavior and engine health to improve fuel efficiency and safety.

Mobile Computing

Mobile Computing is a technology that allows for the transmission of data, voice, and video via a computer or any other wireless-enabled device without having to be connected to a fixed physical link.

Components of Mobile Computing

Mobile Hardware: The physical devices used to access services. This includes smartphones, tablets, laptops, and increasingly, wearables (smartwatches, AR glasses) and IoT sensors.

Mobile Software: The operating systems (iOS, Android) and applications designed to handle the constraints of mobile devices, such as touch interfaces and limited screen real estate.

Mobile Communication: The infrastructure that facilitates wireless data transfer. This includes:

- **Cellular Networks:** 4G LTE, 5G, and the emerging 6G protocols.
- **Short-range Wireless:** Wi-Fi 6E/7, Bluetooth 5.4+, and Ultra-Wideband (UWB).

Features of Mobile computing

- ➔ **Mobility:** Devices must be lightweight and battery-powered to allow movement.
- ➔ **Connectivity:** The ability to maintain a network connection (or gracefully handle disconnections) while moving between different access points or "cells."
- ➔ **Social Interactivity:** Facilitating real-time collaboration and data exchange between users.
- ➔ **Personalisation:** Personalization of the device to the specific user's context, location, and preferences.

Advantages of Mobile Computing

- ➔ It provides unparalleled portability by allowing users to carry powerful digital tools in a compact form factor.
- ➔ It benefits productivity by enabling the completion of professional tasks during transit or away from a traditional office.
- ➔ It provides real-time data access which ensures that the most current information is available instantly from any location.
- ➔ It benefits global connectivity by facilitating seamless communication through high-quality voice, video, and instant messaging services.
- ➔ It provides location-based services that offer highly personalized information and navigation assistance based on precise geographic coordinates.

- ➔ It benefits remote work flexibility by allowing employees to perform their duties effectively from virtually any environment.
- ➔ It provides critical emergency assistance by allowing for immediate contact with first responders and real-time location sharing.
- ➔ It benefits the user's finances by providing significant cost efficiency through the consolidation of multiple electronics into a single device.
- ➔ It provides social integration by keeping individuals connected to their personal communities and professional networks regardless of distance.
- ➔ It benefits data management by ensuring that massive amounts of information remain synchronized and accessible through cloud integration.

Disadvantages of Mobile Computing

- ➔ It creates significant security risks because mobile devices are easily lost or stolen compared to stationary hardware.
- ➔ It enables hackers to use unencrypted public Wi-Fi networks to intercept or steal sensitive data.
- ➔ It suffers from limited battery life which restricts usage time and requires constant access to power sources.
- ➔ It encounters connectivity failures in remote areas or dead zones where signal strength is insufficient.
- ➔ It involves a high risk of physical damage due to the fragile nature of portable screens and components.
- ➔ It causes frequent user distraction by providing a constant stream of notifications that interrupt focus.
- ➔ It eliminates the boundary between professional and personal life because of constant digital accessibility.
- ➔ It demands high financial investment due to expensive data plans and the need for frequent hardware upgrades.
- ➔ It faces inherent resource constraints as mobile processors often lack the power for heavy computational tasks.
- ➔ It promotes digital fatigue and eye strain by encouraging excessive time spent looking at small screens.